Handheld UVGI surface decontamination devices (Courtesy: American Ultraviolet/CureUV)

bilities to emit warning if a human being steps into the room when the UVC lamp is on to prevent any form of injury. Since 2020, we have witnessed introduction of many mobile robots that conduct UVC based sanitisation in the absence of humans. Many of them come with the 'autonomous movement' capability so that they can sanitise the space without human intervention.

Decontaminating the air

As mentioned earlier, inhalation of air carrying fine droplets of infectious virus are now identified as the primary cause for the spread of coronavirus, including the Delta variant. The risk varies with respect to the amount of virus (*aka* virus load) a person is exposed to. Enclosed spaces with inadequate ventilation or air circulation, combined with prolonged exposure, increase the risk of virus transmission exponentially.

CDC recommends that the entire air of a room must be refreshed twelve times every hour, that is, the entire room's air refreshed every five minutes. This rate of refreshing the air reduces the viral load to levels where chances of transmission become negligible. For details, refer to *https://www.cdc.gov/coronavirus/2019-ncov/hcp/infection-control-recommendations.html*

Historically, UVC was already being used to decontaminate air for other pathogens. Let us see how effective different solutions already in play are, and whether any customisations can make them more effective.

Solutions for centralised ACs and AHUs. Centralised air-conditioning is obviously a prime candidate for trouble. While high-efficiency particulate air (HEPA) filters are in place since quite some time, technology improvements have led to the installation of UVGI systems in the air handling units (AHUs). Herein, UVC lamps are installed at the site of condenser coils, the breeding place for germs due to moist condensing water and dust.

Irradiating the cooling coils with UVC radiation ensures there is no growth of germs, dust accumulation, and sterilisation of pathogens within the time it passes through the coils. This solution helps to prevent the spread of airborne disease from one room to another at a low cost. But in most cases this setup cannot prevent localised in-room transmission of infection, as the rate of refreshing of air is not as high as twelve refreshers per hour. A great exception is a typical aircraft, which has AHUs that refresh air at much faster rate, thus eliminating the risk of coronavirus to a considerable degree.

A remote controller UVGI system

In-duct solutions. In-duct UVGI systems are installed within a heating, ventilation, and air-conditioning (HVAC) system. These devices are installed in the outlet duct for rooms and disinfect with a high power dose of UVC. A special treatment of the UVGI keeps the HVAC coils, drain pans, and moist surfaces free of pathogens and produces relatively low levels of UV energy throughout the day.

Coil treatment UVGIs is not recommended for disinfecting air but can be used in HVAC systems to improve operational efficiency. Air-disinfection UVGI installed within the HVAC duct can be effective in disinfecting, as it generally requires more powerful UV lamps. This has better disinfection, particularly if the air is recirculated in the room.

Air purifiers. This is one product category that has seen a large jump in sales since the onset of coronavirus in early 2020. While many are buying non-UV powered air purifiers with the wrong assumption that they will kill coronavirus too, some are buying newly launched UVC integrated units. These units come with UVC lamps installed near their suction blowers so that when placed in a room they can decontaminate the air that passes.

But there are a few risks here. First, the power of the UVC lamp must be adequate to kill the virus within the limited time it will have to disinfect the air. Second, very few have the blower powerful enough to refresh the room twelve times an hour. Hence, they may not reduce the viral load to safe levels, except for very small rooms. Thus, while they will provide an improvement over conventional (HEPA-filter) air purifiers, if adequate safety measures are not maintained the virus may still spread despite the investment!

Lower-air disinfectants. Over time, pathogens tend to settle downwards, and get accumulated on the